

{DALAS} Drug Repurposing Project

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The goal of this project is to predict the success of potential clinical trials for new drug-disease combinations within the family of immune system diseases focusing on drugs used to treat at least one member of this family. We collected and merged data on drugs (physico-chemical properties, molecular fingerprints and molecular targets), diseases (primarily molecules involved in pathogenesis) and connections drugs-diseases (conducted clinical trials and their success, as well as features generated from analysis of shared pathways). The numerical features are standardized to avoid scale bias, and categorical features are encoded as binary (0/1). The final drug dataframe is of dimensions 767×37 , the disease dataframe is 66×10 and the dataframe connecting clinically tested drug-disease combinations has 48 columns. The number of trials varies from 250 to 1500, depending on the strictness of our negative label assignment.

Prediction target is the binary outcome of the clinical trial (success or failure), represented as a single column of the dataframe linking drugs and diseases. Assigning failures is non-trivial, since while successful late-phase results are usually published, early phase trials often lack published results making it unclear whether a trial truly failed (i.e. drug was ineffective) or was successful yet unpublished (or published elsewhere). If we are the most stringent with assignment of negative labels, we get 27 failures and 239 successes, and if the least stringent (assuming failure unless success is shown) - 1131 fails and 239 successes. So we'll need to dig more into how to assign labels better, since it's the crucial point for model training, given that a priori we would expect to see more failures than successes, but failed experiments are not documented as good as successful ones.

To split the data into training and test sets, we will use the corresponding function from scikit-learn module, stratifying by label and setting the test size to 20%. For further validation we plan to use 5-fold cross validation. We plan also taking into account timestamps of corresponding trial successes so that the model is trained on older and tested / validated on more recent data to avoid data leakage. Given a small volume of training data, cross validation will probably yield more accurate and robust error estimates rather than final estimation on a single test dataset.

We envisage using several classification models directly on the merged feature matrix, including SVM and logistic regression as baseline models, random forest and XGBoost to capture non-linearities (potentially experimenting with kernels for SVM as well), particularly leveraging features derived from molecular pathways shared by drugs and diseases. Some of those models will require hyperparameter tuning (SVM kernel choice, random forest `n_estimators`...). In addition to this, we want to directly use raw vectors of pathway IDs associated with each disease and drug in graph-based models, since the natural way to model this data is as a bipartite drug-disease graph

with an additional layer of pathway nodes. Although Graph Neural Network would require more training data. If we realize that we have too few data points for robust model training, we can further broaden the scope of diseases considered. Also since we are sure only of positive labels, we can adapt semi-supervised models such as positive-unlabeled (PU) approaches treating initially all unknown labels as negative and then iteratively adjusting labels based on the data structure (proximity to “surely positive” points). Finally, to increase the interpretability, we plan to obtain, where possible, feature importance metrics.

During model training and evaluation we will need to ensure class balance (e.g. using SMOTE or class weights) and report not only accuracy, but also precision, recall, F1-score as well as ROC-AUC / PR-AUC (for models yielding probabilities such as logistic regression) to make sure that minority class is not neglected.

Of note, we decided to exclude cosine similarity of drug and disease name embeddings in PubMedBERT since this language model was trained on a present corpus of scientific articles, which poses the problem of data leakage and unreliable evaluation / results since newly discovered successful combinations will eventually co-appear a lot in new research. This approach is still interesting for predicting the success of potential new studies. However, to properly include it, we would need language models trained only on research published before each trial’s results, which may be excessive for this project.